

Intercepts



1

- | | | | | | |
|----------|------------------|-------------------|----------|--------------------|-------------------|
| a | i $x = 0$ | ii $y = 0$ | b | i $x = 3$ | ii $y = 4$ |
| c | i $x = 0$ | ii $y = 0$ | d | i $x = 4.5$ | ii -3.5 |

2

- | | |
|--------------------|-------------------|
| a $(0, -5)$ | b $(0, 3)$ |
|--------------------|-------------------|

3

- a** $y = 3$
b The y -intercept is the constant without x in the equation.
c $y = c$

4

- a** The constant term of 4 is the same.
b All the graphs intersect the y -axis at the point $(0, 4)$.
c The line will pass through $(0, 4)$.

5

- | | | | |
|------------------|-------------------|----------------------------|----------------------------|
| a $y = 3$ | b $y = -5$ | c $y = -3$ | d $y = 4$ |
| e $y = 0$ | f $y = 2$ | g $y = \frac{2}{3}$ | h $y = \frac{8}{5}$ |

6

- | | | | |
|--------------|--------------|--------------|--------------|
| a Yes | b Yes | c No | d No |
| e Yes | f No | g Yes | h Yes |

7

- | | | | |
|------------------|-------------------|----------------------------|-------------------|
| a $x = 1$ | b $x = -1$ | c $x = 2$ | d $x = -3$ |
| e $x = 0$ | f $x = 2$ | g $x = \frac{1}{3}$ | h $x = 2$ |

8

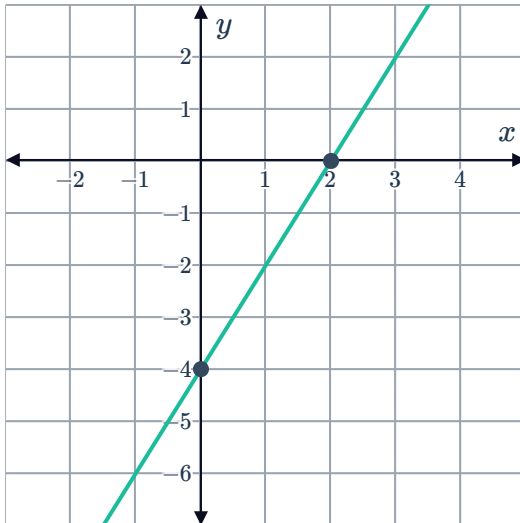


a

i $(0, -4)$

ii $(2, 0)$

iii

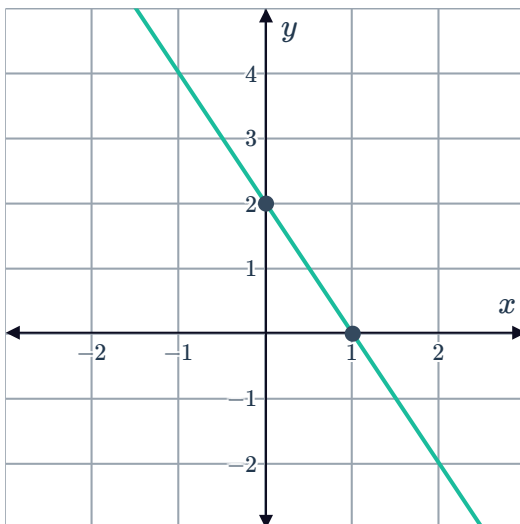


b

i $(0, 2)$

ii $(1, 0)$

iii



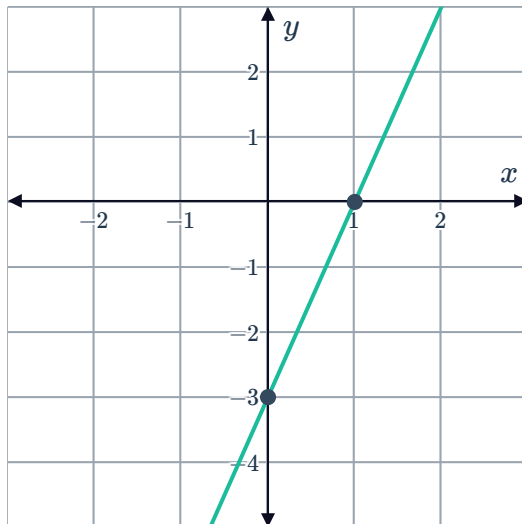
c

i $(0, -3)$



ii $(1, 0)$

iii

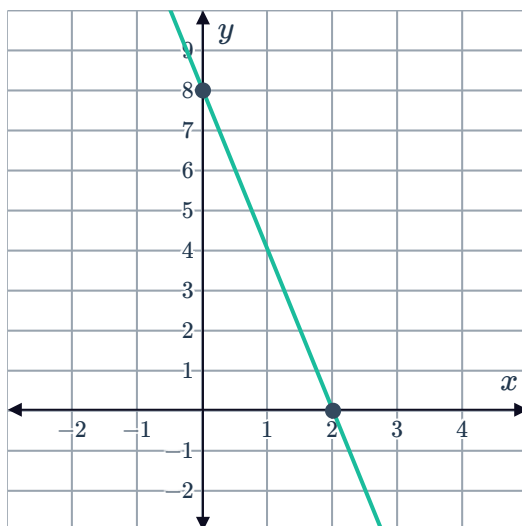


d

i $(0, 8)$

ii $(2, 0)$

iii



Lines through the origin

9

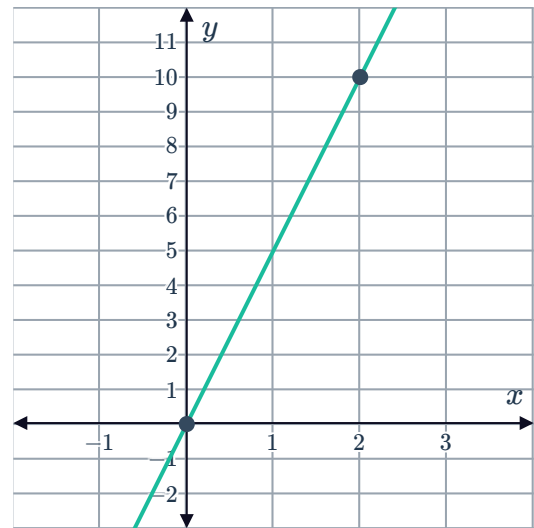
a $(0, 0)$

b $(0, 0)$

c $y = 10$



d

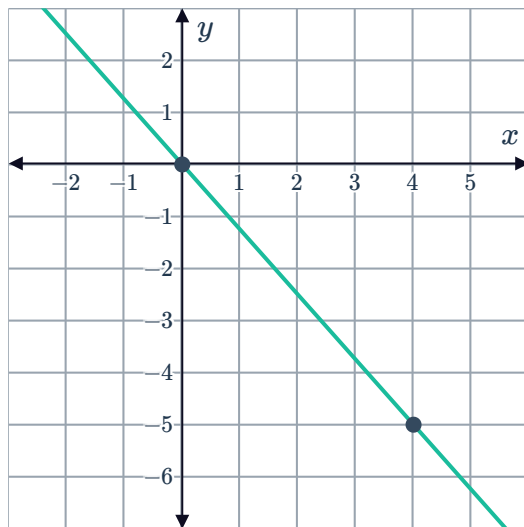


10

a $(0, 0)$

b $(4, -5)$

c



11 There is no constant term as the value of c is zero.

12

a No

b Yes

c Yes

d Yes

e Yes

f Yes

g Yes

h No

Gradient

13

a $y = 4$

b $y = 6$

c 2

d $m = 2$

e It is the coefficient (number in front) of



14

- a The coefficient of x is the same, 4.
c The line will have gradient of 4.

- b All the graphs have gradient of 4.

15

a $m = 9$

b $m = -7$

c $m = \frac{5}{4}$

d $m = -1$

16 $y = 2 + 7x, y = 7x, y = 7x - 2$

17

a

i $m = 2$

ii $c = 9$

b

i $m = 5$

ii $c = -6$

c

i $m = -5$

ii $c = 8$

d

i $m = -4$

ii $c = -2$

e

i $m = 4$

ii $c = -\frac{1}{2}$

f

i $m = -2$

ii $c = -\frac{2}{3}$

g

i $m = 5$

ii $c = 4$

h

i $m = \frac{3}{2}$

ii $c = 3$

18

a $y = 2x + 5$

b $y = -3x + 2$

c $y = -2x - 1$

d $y = 4x$

e $y = -7$

f $y = 4$

g $y = \frac{1}{2}x - 2$

h $y = -\frac{3}{4}x + \frac{1}{2}$

Approximate solutions of equations



19

a $x \approx 1.2$

b $x \approx -6.7$



20

a $x \approx 2.3$

b $x \approx 2.3$

c $x \approx 0.3$

d $x \approx 0.3$

21

a $x \approx 2.2$

b $x \approx 2.3$

c $x \approx 1.5$

d $x \approx 0.6$

e $x \approx 2.9$

f $x \approx 13.3$

g $x \approx -0.1$

h $x \approx -10.5$